

**Indian Institute of Information Technology, Bhopal****Department of CSE /ECE**

Branch: CSE/ECE

Semester: 2<sup>nd</sup>

Section:

**Mid-Term Examination 13-03-2026 (Friday)****Subject Code & Name:** CSE/ECE -2005,  
Object Oriented Programming

Maximum Marks: 20

Time Allotted: 90 Minutes

- All questions are compulsory. Write your answers clearly and legibly.
- While solving, clearly mention the question numbers and their respective parts.
- Attempt the questions in the given sequence. All subparts of a question must be answered together in the same place.
- If any question appears ambiguous, do not worry; clearly state your assumptions in capital letters before answering.
- While writing programs, use correct syntax as required in a code editor for successful compilation and execution. Only fully correct programs without syntax errors will be awarded full mark

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| 1 | A | What is the purpose of a friend class in C++? Explain briefly with proper syntax and a suitable example.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1 |
|   | B | A company wants to design a class Box that stores length, breadth, and height as private data members. The company requires a function to calculate the total volume when two Box objects are combined. This function must not be a member of the class, but it must be able to access the private data members of both objects. Write a complete C++ program with proper syntax that compiles successfully on a standard C++ compiler. Also determine the final output.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2 |
|   | C | A software developer wants to design a class named MyTime that stores hours, minutes, and seconds as private data members. The first object must be initialized with the values 2 hours, 45 minutes, and 50 seconds, and the second object must be initialized with 1 hour, 20 minutes, and 30 seconds. The data members must be initialized strictly through a parameterized constructor, and no separate member function should be used for initialization. The class must support addition of two MyTime objects using the + operator. While performing addition, if the sum of seconds is 60 or more, it must be converted into minutes, and if the sum of minutes is 60 or more, it must be converted into hours. The final result must be displayed in the format hh:mm:ss using a normal member function. Write a complete C++ program with proper syntax that compiles successfully on a standard C++ compiler. Also determine the final output. | 2 |
| 2 | A | What is a destructor in C++? Explain its purpose, provide the general syntax, and illustrate it with a suitable example program.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1 |

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|           | <b>B</b> | A software developer wants to design a class named MyMatrix that represents a 2 by 2 matrix. The matrix elements must be stored as private data members. The class should provide a parameterized constructor to initialize the MyMatrix values. A member function named add should be implemented that takes another MyMatrix object as an argument and returns a new MyMatrix object representing the sum of the two matrices. The program must demonstrate the concept of passing objects as arguments and returning objects from functions. The result of the matrix addition should be displayed using a normal member function. Write a complete C++ program with proper syntax that compiles successfully on a standard C++ compiler. Also determine the final output when the first matrix is initialized with values 1 2 3 4 and the second matrix is initialized with values 5 6 7 8.                                                                                                                                                                                   | <b>2</b> |
|           | <b>C</b> | Write a C++ program that defines a class named Printer to demonstrate function overloading. The class should contain four overloaded member functions named print: one that accepts a single integer parameter, one that accepts a single double parameter, one that accepts a C style string void print(const char* str), and one that accepts two integer parameters where the second parameter has a default value. Each overloaded function must print a message clearly indicating the data type or parameter combination it corresponds to, such as "int version", "double version", "const char* version", or "int, int version". In the main() function, create an object of the class and invoke the print function using the following calls: p.print(10);, p.print(10.5);, p.print('A');, p.print("Hello");, and p.print(5, 6);. For each function call, determine which overloaded function is selected by the compiler and justify your answer using C++ overload resolution rules. Clearly state whether any call results in ambiguity, and explain why or why not. | <b>2</b> |
| <b>3.</b> | <b>A</b> | What is the copy constructor ? Briefly explain with syntax and an example.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>1</b> |
|           | <b>B</b> | Explain the static keyword in C++. Write the syntax to declare and initialize a static data member of a class.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>1</b> |
|           | <b>C</b> | What is the expected behaviour of the program? Please show calculation and write your output clearly.<br><pre>#include &lt;iostream&gt; using namespace std; int x = 10; void show() {     static int x = ::x + 5;     cout &lt;&lt; x &lt;&lt; " ";     x = x + 2; } int main() {     int x = 20;     {         int x = 30;         cout &lt;&lt; x &lt;&lt; " ";         cout &lt;&lt; ::x &lt;&lt; " ";         show();         cout &lt;&lt; endl;     }     cout &lt;&lt; x &lt;&lt; " ";</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>1</b> |

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|     |     | <pre> cout &lt;&lt; ::x &lt;&lt; " "; show(); cout &lt;&lt; endl; return 0; } </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                               |
| C   | 2   | <p>Write a C++ program to demonstrate the use of a copy constructor in a simple class. Design a class named Employee with private data members empId (int), name (string), and salary (float). Provide a parameterized constructor to initialize these data members. Implement a copy constructor that accepts its argument as a constant reference to an existing object of the same class. Include a member function display() to print the employee details. In the main() function, create one object using the parameterized constructor and then create a second object by copying the first object. Modify the name of the second object and display the details of both objects to verify that the original object remains unchanged. Also identify the statement at which the copy constructor is invoked.</p> |                                                                                                                                                                                                                                               |
| 4   | 5   | <p>Please read the following programs carefully and write the output or expected behavior. Show all necessary calculations wherever required.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                               |
| 4.1 | 4.2 | <p>(1 Mark)</p> <pre> #include &lt;iostream&gt; using namespace std; int main() {     int x = 2;     int *p = &amp;x;     int **q = &amp;p;     int ***r = &amp;q;     ***r = ***r * 3;     **q = **q + 4;     *p = *p - 1;     cout &lt;&lt; x &lt;&lt; endl; } </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>(1 Mark)</p> <pre> #include &lt;iostream&gt; using namespace std; namespace A {     int x = 5; } namespace B {     int x = 10; } using namespace A; using namespace B; int x = 5; int main() {     cout &lt;&lt; x &lt;&lt; endl; } </pre> |
| 4.3 | 4.4 | <p>(0.5 Mark)</p> <pre> #include &lt;iostream&gt; using namespace std; int main() {     int a = 5, b = 10;     int *q = &amp;a;     const int *p = q;     q = &amp;b;     cout &lt;&lt; *p &lt;&lt; " " &lt;&lt; *q; } </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <p>(0.5 Mark)</p> <pre> #include &lt;iostream&gt; using namespace std; int main() {     int a = 5, b = 10;     int *q = &amp;a;     const int *p = q;     q = &amp;b;     *p = 20;     cout &lt;&lt; *p &lt;&lt; " " &lt;&lt; *q; } </pre>    |
| 4.4 | 4.5 | <p>(0.5 Mark)</p> <pre> #include &lt;iostream&gt; using namespace std; int main() {     int a = 5, b = 10;     int * const p = &amp;a;     *p = 20; } </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p>(0.5 Mark)</p> <pre> #include &lt;iostream&gt; using namespace std; int main() {     int a = 5, b = 10;     int * const p = &amp;a;     p = &amp;b; } </pre>                                                                               |

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|     | <pre>         cout &lt;&lt;*p &lt;&lt;" "&lt;&lt;a&lt;&lt;" "&lt;&lt;b&lt;&lt; endl;         return 0;     } </pre>                                                                                                                                                                                                                                                                                                   |  | <pre>         cout &lt;&lt;*p &lt;&lt;" "&lt;&lt;a&lt;&lt;" "&lt;&lt;b&lt;&lt; endl;         return 0;     } </pre> |
| 4.6 | <p>(1 Mark)</p> <pre> #include &lt;iostream&gt; using namespace std; int update(int &amp;r, int val = 5) {     r += val;     val += 10;     return r + val; } int main() {     int a = 10;     int b = 20;     int &amp;ref = a;     int x = update(ref);     int y = update(ref, b);     cout &lt;&lt; a &lt;&lt; " " &lt;&lt; b &lt;&lt; " "&lt;&lt; x &lt;&lt; " " &lt;&lt; y &lt;&lt;endl;     return 0; } </pre> |  |                                                                                                                     |